Analysis of Algorithms

Project – Presentation I

Students enrolled in *Analysis of Algorithms* are expected to work on a project during the semester, which will be 30% of the final grade, split between two different terms. There are two presentations of the project. In the first presentation, the students present the first version of the algorithm, the expectation is to have a complete project including the implementation and the analysis. In the second presentation, the student presents new results, updated optimizations, and addresses any comments from the first version.

What is the project about?

Your job is to become the class's expert on a given algorithm, identify a meaningful problem to which it can be applied, understand its theoretical properties, formulate questions about its behavior, design experiments to answer those questions, compare it against appropriate alternatives, and defend their conclusions. Each person is assigned an algorithm, look at the assignments sheet.

Project Presentation I (Implementation and Preliminary Results)

1. 12-15 page report using IEEE Conference format (https://www.ieee.org/conferences/publishing/templates).
2. Oral presentation (10 minutes presentation, the first presentation is recorded.)
3. You have to upload to Moodle the following files:
    1. PDF file of the report.
        1. In the Annex section, include: 1) a link to Overleaf for the manuscript, 2) a link to GitHub with the implementation of the algorithm, 3) link to the video presentation.
    2. Slides of the presentation as a PDF file.
4. The report and the presentation have to include the following aspects (at a minimum):
    1. Name of the selected algorithm.
    2. Why is it important?

3. What problem(s) it solves?
4. Explanation of the algorithm.
5. Implementation details. Use C++ for the implementation.
6. Results
7. Conclusions

5. Check the class schedule for information about the deadline.